



9 A programmer is writing a program using an integrated development environment (IDE).

(a) The program will take data from bar codes as input.

A bar code includes a check digit.

Explain how the check digit is used to check for errors in data entry.

.....

.....

.....

.....

.....

..... [3]

(b) The programmer is using a high-level language to write the program.

(i) Describe what is meant by a high-level language.

.....

.....

.....

..... [2]

(ii) The programmer uses a compiler during the development of the program.

Describe the benefits of using a compiler when developing the program.

.....

.....

.....

.....

.....

..... [3]

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (Cambridge University Press & Assessment) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in our Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge International Education is the name of our awarding body and a part of Cambridge University Press & Assessment, which is a department of the University of Cambridge.

